# Basic Electronics

# 14


Have you ever wondered how a light turns on when we flip a switch? Or how a push of a remote-control button turns on the TV? These happen because of electricity. Electricity plays a vital role in the modern world, enabling us to do a variety of tasks. In this lesson, we will talk about electricity, what constitutes electricity, and how it flows in electrical circuits.

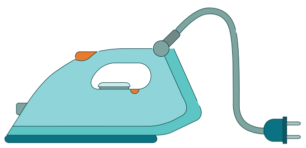
# 14. Basic Electronics

## Electricity

Electricity is a form of energy that flows through materials and allows us to do simple chores. It is used to power electrical items like televisions, lights, electric irons, computers etc.

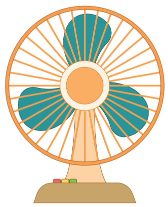
For example:

**a.** Heating - Electric Iron



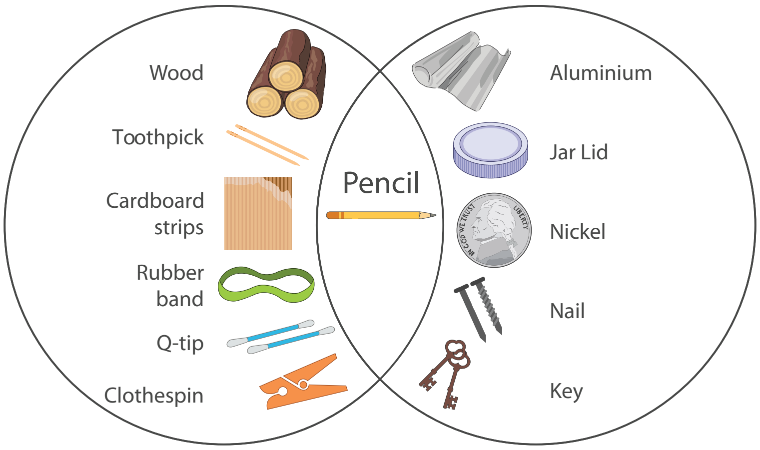
**b.** Lighting - LED lights

**c.** Power Machines - Fan



We require electricity to run all the household items shown in the above examples. For this to happen, electricity must flow from one point to another. There are a few materials which can conduct electricity from one place to another. The materials that have the ability to conduct electricity are called conductors.

For example, aluminium or copper wires are conductors since they allow electricity to flow through them. Materials that do not allow electricity to flow through them are called insulators.




## How electricity works

Four concepts combine to make electricity work:

**a.** Electrons

**b.** Electric Current

**c.** Voltage

**d.** Resistance

Everything in the universe is made of tiny particles called atoms. Each atom contains even tinier particles called protons, neutrons, and electrons. The centre of an atom is called the nucleus. It is made of protons and neutrons. The electrons revolve around the nucleus similar to how planets orbit the Sun.

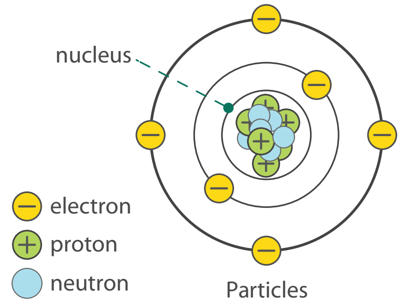
## Atom, neutron, proton, and electron

**a.** An atom is the basic building block for all the matter in the universe. The size of an atom is extremely tiny.

**b.** A proton is a part of an atom that is positively charged.

**c.** An electron is a part of an atom that is negatively charged.

**d.** Neutrons do not have a positive or negative charge, making them neutral particles.



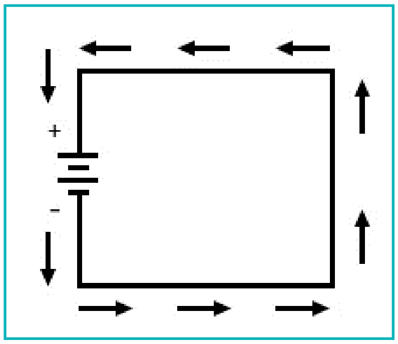
Particles

The positive and negative charges of protons and electrons act like opposite sides of a magnet. They attract each other, resulting in an electric current.

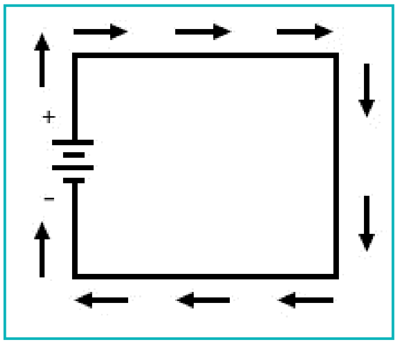
## Electric Current

When energy is applied to a conductor, electrons start to move from one atom to the next. This movement of electrons generates an electric current. Therefore, electric current is expressed by the amount of charge flowing through a particular area per unit of time. It is measured in Amperes (A).

Electric current flows in a positive to negative direction. This is opposite to the direction that electrons flow in. (negative to positive).

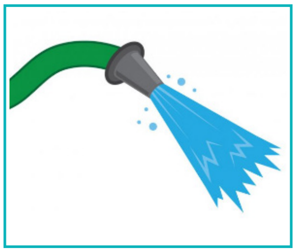


Flow of electrons



Flow of electric current

We are familiar with the word water current. We know that flowing water constitutes a water current in rivers or in a water pipe. We hear people saying, “This pipe has a strong water current”. That means there is a large amount of water moving through the pipe.



Similarly, a strong electric current means that there are a large number of electrons flowing through a conductor. The flow of electrons depends on the energy source which forces them to move.

## Energy sources

Energy is “the ability to do work”. It takes energy to cook food, to drive to school, and to jump up in the air. Similarly, to move electrons, energy is required. That energy can be generated by various energy sources. These energy sources are categorised into two main types:

### 1. Renewable energy sources

A source of energy is considered renewable if it comes from natural sources and can renew itself over time. For example:

**a.** solar power generated by the sun

**b.** wind power generated by wind turbines

**c.** waterpower (sometimes called hydropower) generated by running or falling water

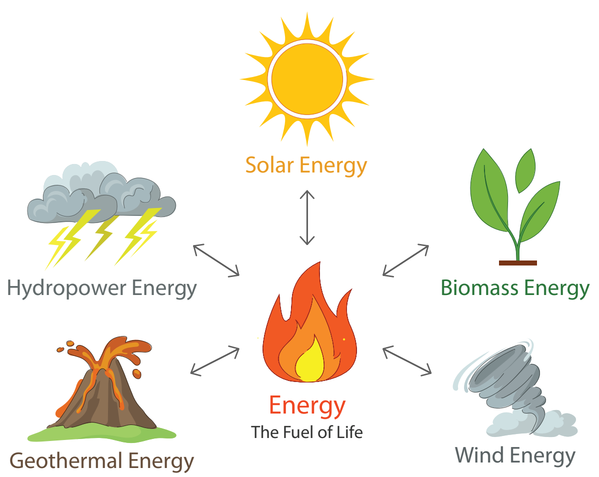
**d.** geothermal energy from the earth

**e.** biomass energy from organic materials

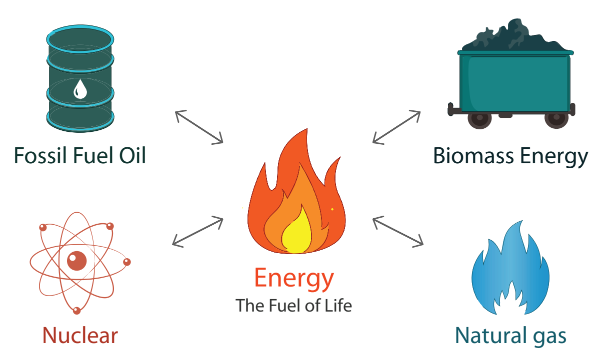
### 2. Non-renewable energy sources

Non-renewable resources are natural resources that exist in fixed amounts and can be completely used up. For example:

- burning fossil fuels (oil, gas, coal) at power stations. These fuels are formed over hundreds of millions of years from the remains of plants.




Renewable sources




Non-renewable sources

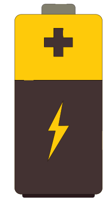
Energy derived from energy sources can be stored and used as required. A device used to store energy is called a battery.

**Bytes**

Energy can neither be created nor be destroyed. It can only be transformed from one form to another.

## Battery

A battery is a sort of container that stores energy until it is completely used up. Chemicals inside the battery store energy. When the battery is used, chemical energy is then converted into electrical energy.

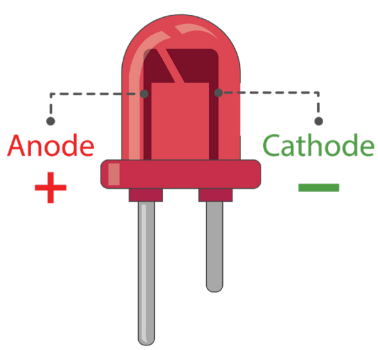


## LED

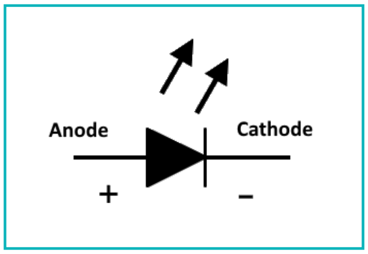
A Light Emitting Diode (LED) is an electronic component that releases one colour of light when an electric current pass through it in the expected direction. (Positive to negative)

An LED has two terminals, **Anode** which is positive and **Cathode** which is negative.

To identify positive and negative terminals in an LED is very simple, the longer leg represents the Anode (positive), and the shorter leg represents the Cathode (negative). When we look inside an LED, we see two metal portions adjacent to each other, the LED terminal connected to the bigger portion is the Cathode and the terminal connected to the smaller portion is the Anode.




**Symbol:**




## Exercise

### Fill in the blanks

1. Materials that conduct electricity are called __________________. Materials that cannot conduct electricity are called __________________.

2. Electric current is measured in __________________ .

3. An LED has two terminals, __________________ is positive and __________________ is negative.

## Healthy Practices:

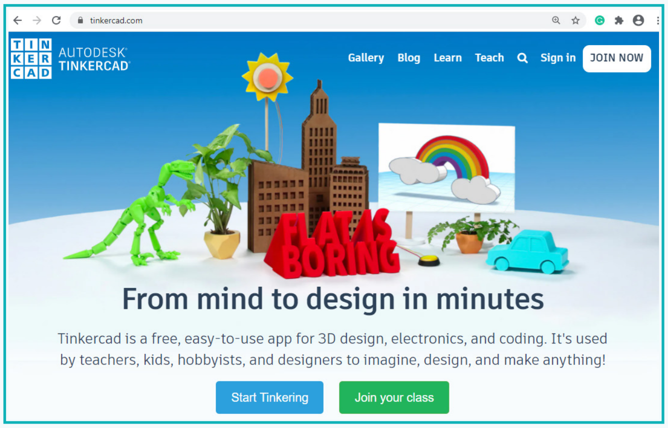
- Never stick your fingers or any object into an electrical outlet or a light bulb socket.
- Keep fingers and other objects out of small appliances, such as toasters, even if the appliance is off.
- Never use an appliance near a sink, bathtub, or any other source of water.
- Keep electrical wires and appliance cords away from sources of heat.
- Never touch an electrical appliance or device, such as a light switch, hair dryer or toaster, if you are touching water.

## Activity

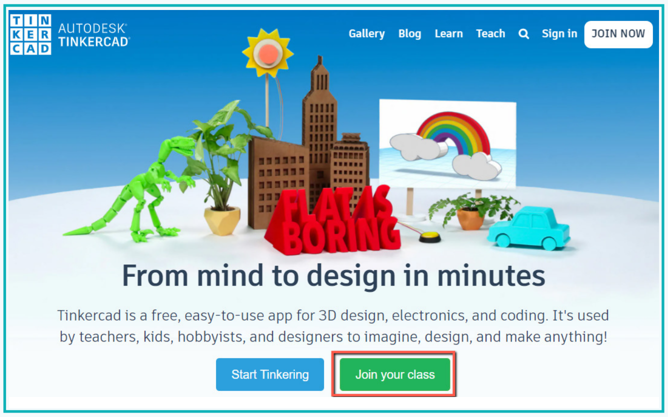
Use TinkerCad to design a virtual electric circuit.

We will create a virtual electric circuit instead of a real circuit, because the software allows us to analyse and verify performance before we build a physical prototype. Virtual modelling solves real-world problems safely and efficiently.

1. Open tinkercad.com in a browser.

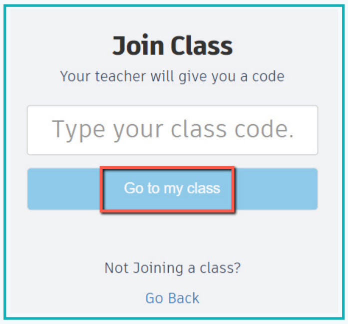



2. Click on “Join your class”

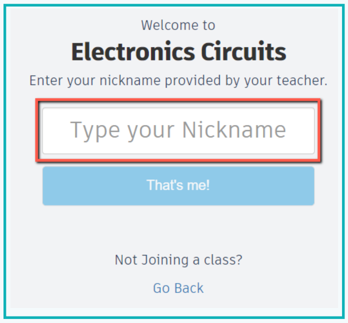


## Activity

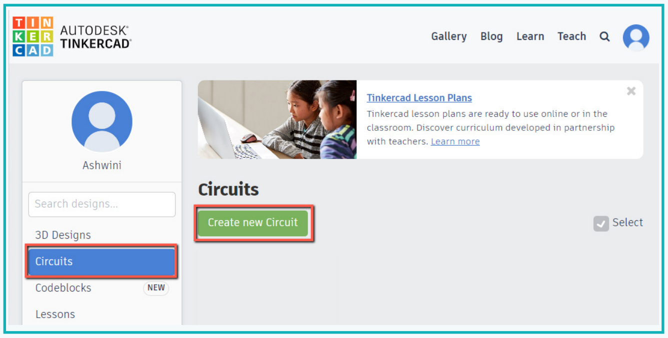
**3** Get a code from your teacher and click on “Go to my class”

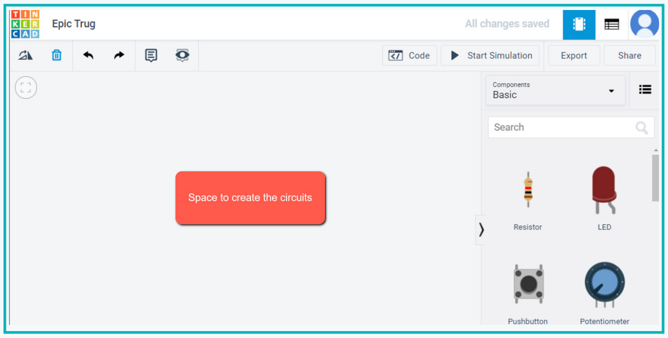



**4** Enter your name

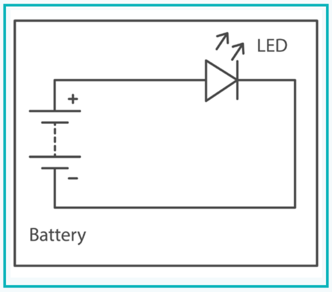



**5** To create a new circuit, click on “create new circuit”

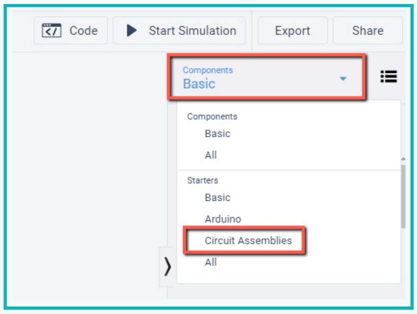


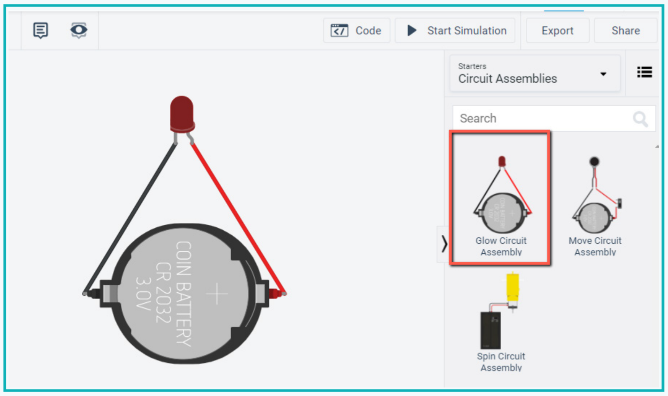

**Circuit:** A circuit is a path which allows electricity to flow through it. A circuit needs a battery (which works as a source of electricity) and materials (which allow the electricity easily).



## Activity




6. To check prebuilt circuits, select "circuit assemblies" Under the Starter section.




7. Drag and drop the "Glow Circuit Assembly"

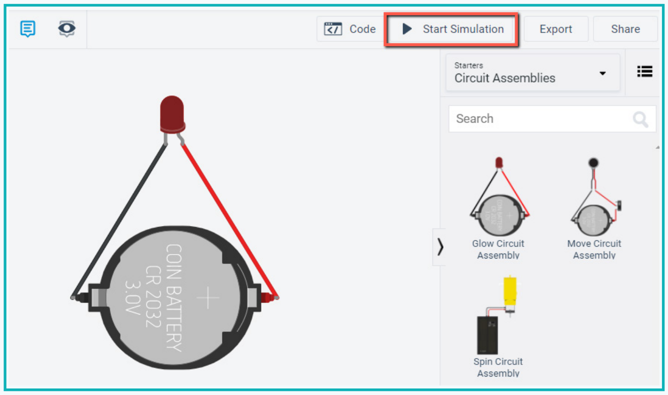
**a.** This circuit is made using a coin cell 3v battery and an LED.

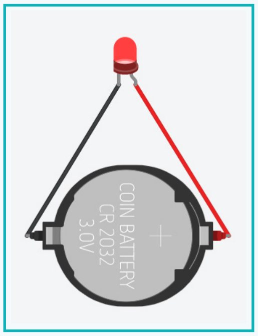
**b.** The coin cell 3V battery will be used as a source of energy. This battery is used in watches.

**c.** The LED is small light which lights up when electricity flows through it.

Here, the negative point of the LED is connected with the negative point of the coin cell battery. The positive point of the LED is connected to the positive point of the coin cell battery.

8. To check the circuit, click on "start simulation" and LED will be ON.




LED is ON

9. Click on "stop simulation" and the LED will go OFF

## Exercise

**State whether True or False**

4. An electric current flows from negative to positive direction.

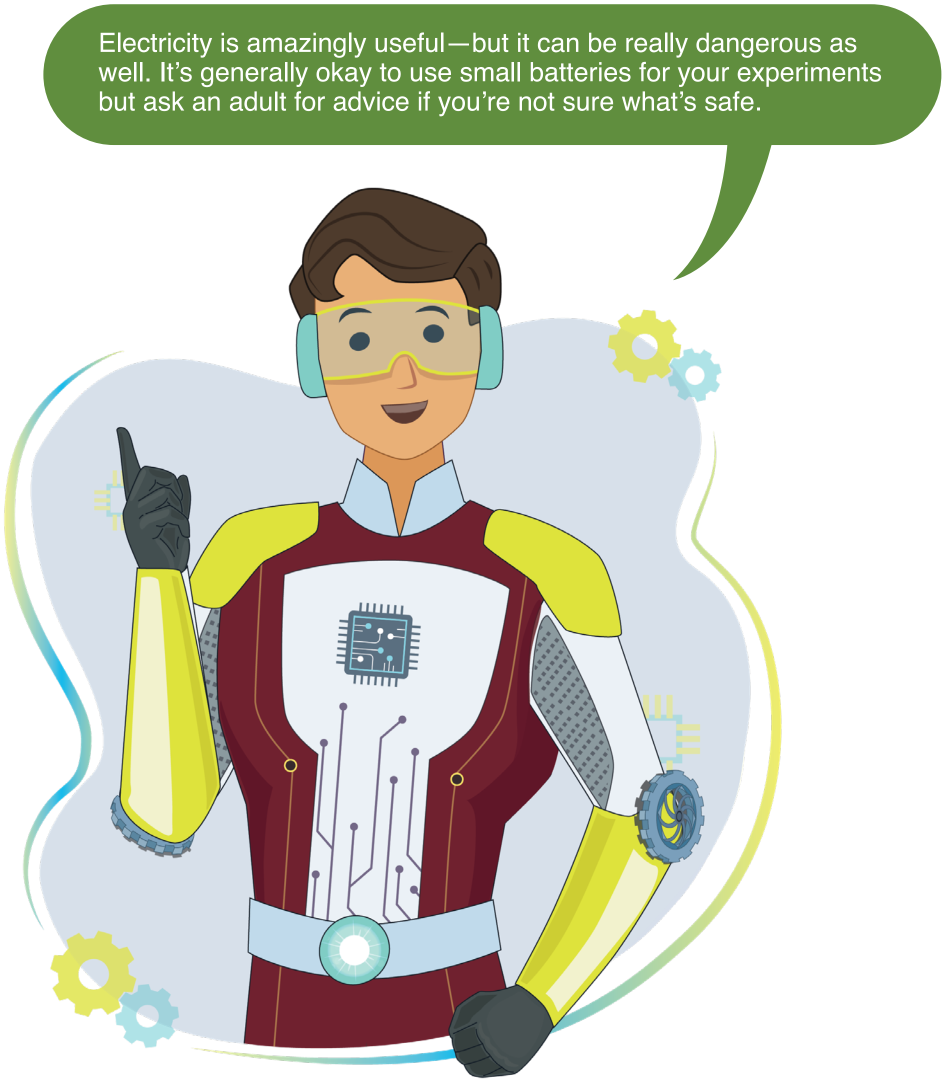
▲ ____________________

5. Non-renewable resources are natural resources that exist in fixed amounts and can be completely used up.

▲ ____________________

Electricity is a form of energy that flows through wires made up of conductors like copper or aluminium. Electricity is a controllable and convenient form of energy for a variety of uses in homes, schools, hospitals, industries and so on. Atoms of metal are made up of electrons which freely move from one atom to the next. Electric current is expressed by the amount of charge flowing through a particular area in unit time. Today, electricity provides most of the energy which is used to run the modern world.

ICT 360




## Tech Flash

- **Electricity** : Electricity is a form of energy that can be carried by wires or any materials.
- **Conductors** : A conductor is a material that conducts electricity.
- **Insulators** : An insulator is a material that cannot conduct electricity.
- **Electric Current** : An electric current is the flow of electrons through a conductor.